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Review on research progress in boiling acoustics

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ABSTRACT

Ever-growing miniaturization of electronic devices and space-conserving endeavours of heat transfer systems pose a challenging task to the current cooling strategies. Boiling acoustics is one of the most potent and efficacious methodologies to reliably predict the boiling regime inside the cooling systems to assay the safety and address the emergency conditions. Boiling acoustics is gripping attention out of the few foreseeable technologies for the future cooling requirements. Though the potentiality of boiling acoustics was unravelled in the late 1990s, the research data present in this arena is however lacking. This paper presents a comprehensive review of the literature reported from the 1970s to the present date. Furthermore, this paper also details the evolution of boiling acoustics from the initial application of boiling incipience detection to regime identification. Much focus is given on salient features of boiling acoustic characterization dealing with the regime detection. Effects of various parameters such as thermos-physical properties of the heater surface and the boiling liquid that directly or indirectly influence the acoustic spectra are also presented. The prediction of the boiling regime constitutes the first necessary step in producing autonomous cooling systems. Hence, the detection and characterization of boiling noise under various conditions such as pressure, heat flux, and flow rate is essential.

1. Introduction

For the past three decades, the two-phase heat transfer technique has been in the spotlight for its astonishing heat carrying capacity. No practical and foreseeable process is justified its potential to become the viable alternative. The impediments associated with phase change heat transfer cooling systems (specifically boiling systems) are complex to resolve [1–3]. The critical complexities accompanying this research includes understanding the phenomenon by the heat transfer parameters/variables, reliably predicting the heat transfer rate and boiling regime [4]. Although the three mentioned entanglements are distinct on a macro scale, presenting with the theory, they are interlinked. Many researchers tried establishing a relationship between boiling heat transfer parameters and cooling rate; their analytical expressions are notorious only for their specific or similar experimental conditions. Those individual predictions failed to adequately meet the reasonable accuracy when applied to boiling conditions of other designs. It is largely due to the dependency of boiling heat transfer on the thermo-physical properties of the heat transfer surface and liquid media [5,6]. Table 1 lists the popular nucleate boiling heat transfer correlations.

Boiling heat transfer enhancements can be classified into active techniques and passive techniques. Among active techniques of boiling heat transfer enhancement, acoustic methods are popular. The acoustic wave facilitates the removal of vapour bubbles from the heater surface by disturbing the contact line of the bubble near the heater surface. This early departure of the vapour bubble reduces the possibility of reaching film boiling and enhances heat transfer to a large extent [11]. The passive techniques are flexible in designing and simple in construction compared to the active techniques. The passive techniques can be further classified into two groups: surface modification techniques and boiling liquid modification techniques. Surface modification techniques involve enhancing surface properties such as surface roughness and texture. On the other hand, boiling liquid modification techniques involve enhancing the liquid's thermal conductivity, viscosity, and surface tension. Micro/nano structural changes (prevalent in aging) on the heater surface portray a crucial role in determining the boiling heat transfer. Besides the convections near the heater surface caused either by structural changes or the bulk fluid motions improves the heat transfer coefficient significantly [12]. Fig. 1 shows the comparison of heat transfer augmentations with surface roughness, coatings and nano fluids.

1.1. Approach to BA

From the boiling curve as shown in Fig. 2 [22], and it is well-known

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Nomenclature

μ_l	Dynamic viscosity
h_{fg}	Latent heat of vaporization of the liquid
g	Acceleration due to gravity
ρ_l	Density of the liquid
ρ_v	Density of the vapour
c_{pl}	Specific heat of liquid
ΔT	Wall superheat
Pr_l	Prandtl number of the liquid
a, b and c	Empirical constants
q	Heat flux
q_0	Reference heat flux
k_l	Thermal conductivity of the liquid
N	Active nucleation site density
R_z, R_a	Surface roughness parameters
R_{z0}, R_{a0}	Reference surface roughness parameters
h	Heat transfer coefficient
h_0	Reference heat transfer coefficient

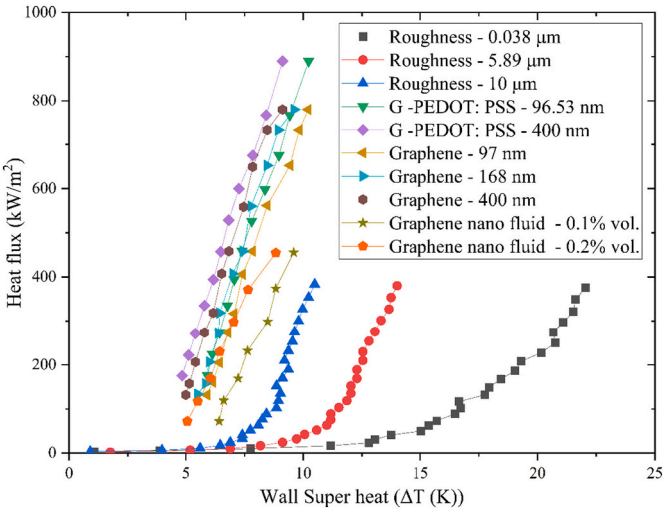


Fig. 1. Comparison of heat transfer augmentations with surface roughness, coatings and Nano fluids [13–21].

that the bubble dynamics vary distinctly among the boiling regimes. To understand these boiling regimes, many researchers in the past employed a photographic method for observation [23]. In contrast, most common cooling system designs may not adapt to the photographic way due to design restrictions. Hence the photographic technique is strictly bounded to laboratory testing cases. In recent times, the research into the Boiling Acoustics arena has been gaining popularity due to its advantages in predicting the boiling heat transfer regime. BA requires no photographic imaging and therefore simple design of boiling chambers would suffice.

A hydrophone probe and/or accelerometer and/or microphone sensor are sufficient. These sensors typically capture the acoustic data sourced from the dynamic bubble behaviour and reliably predict the boiling heat transfer regime [24–27]. Fig. 3 gives the approximate timeline of major events in BA research. The following sections of this review emphasize the key contrast in acoustic data as the boiling regime changes.

Understanding any complex research problem is associated with two specific types of solutions. One of which is predicting the consistent results by the empirical relations. The empirical relations in common are from the results of the specific experiments. Hence, in most cases, empirical results are eminently suitable for that particular experiment or similar experiments under the identical conditions. Another solution approach involves the application of a real-time system, sensing the critical parameters by which observers and artificial neural networks can undoubtedly predict the desired outcome. Boiling Acoustics comes

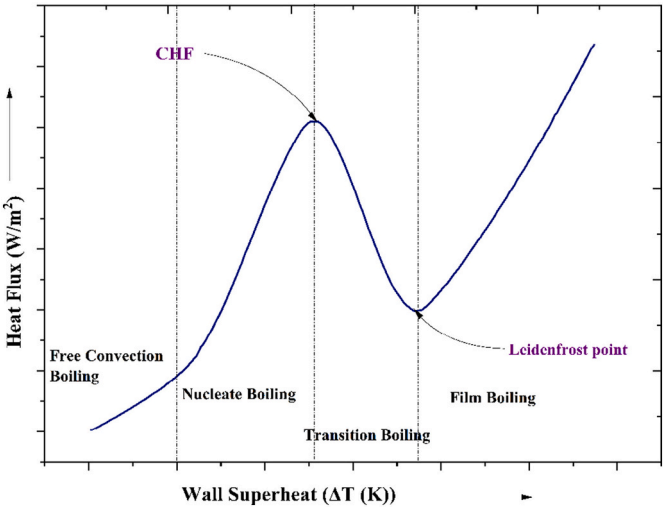


Fig. 2. Pool boiling curve [22].

under the real-time observing system. When the boiling regime goes out of the design plan, the system should either alert the observer by way of an alarm or undertake preventive steps to bring the whole plant to stable operating conditions. The research data available on BA is significantly less. This research needs some focus as it is the only foreseeable

Table 1
Nucleate boiling heat transfer correlations available in the literature.

Authors	Correlation	Reference
Mikic & Rohsenow	$q = \mu_l h_{fg}\sqrt{\frac{g(\rho_l-\rho_l)}{\sigma}}\left[\frac{c_{pl}\Delta T}{c\, h_{fg} Pr_l^b}\right]^a$	[7]
Tien	$q = Ck_l\, Pr_l^{0.33}\, N^{0.5}\Delta T$	[8]
Danilova	$q = c\left(\frac{R_z}{R_{z0}}\right)^{0.2}(0.14 + 2.2\, Pr_l)q^{0.75}$	[9]
	$\frac{h}{h_0} = c\, F(Pr)\left(\frac{q}{q_0}\right)^n ; c = \left(\frac{R_a}{R_{a0}}\right)^{0.133} ; R_{a0} = 0.4\ \mu m$	
Gorenflo	$F(Pr) = 1.73\,(Pr)^{0.27} + \left(6.1 + \frac{0.68}{1-Pr}\right)Pr^2$ and $n = 0.9 - 0.3Pr^{0.15}$ for water	[10]
	$F(Pr) = 1.2\,(Pr)^{0.27} + \left(2.5 + \frac{1}{1-Pr}\right)Pr$ and $n = 0.9 - 0.3Pr^{0.3}$ for other fluids	

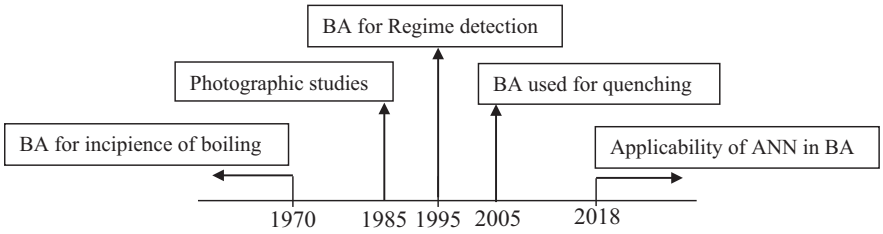


Fig. 3. Approximate timeline of research in boiling acoustics.

technology that heavy power plants can adapt. Hence, the accurate characterization of this acoustic data is of utmost importance.

2. Boiling acoustics (BA)

The review of Anderson et al. [27] in the late 1970s signified that the boiling acoustics is one of the potential methodology to study the boiling regime inside the reactor vessel to assay its condition of emergency. All the established investigations of that time carry three well-classified problems associated with BA approach. These include quality of the reported sound intensities, systems of limited frequency range, and complexity in collating investigation results due to variations in the measurement/analysis methodologies. Also, Bonnet Jr. and Osborn [28] emphasize that uncovering the boiling noise in the background of power reactor noise makes the acoustic boiling detection dubious for several events. Besides specific disadvantages, numerous researchers focused on investigating the noise spectra of boiling. In the boiling acoustic spectrum, resonance-like peaks have been reported by multiple researchers. Out of many classical literature works, the experimental investigations of Nishihara, Nishihara and Bessho [29,30] used cross-correlation to identify resonance patterns that are mostly or at least partly due to the container system resonances. Except at the resonance frequencies, the triangulation [31,32] of these acoustic signals leads to the earlier detection of anomalous boiling regions, thus helps in safe reactor operation. Hence, detection and characterization of boiling noise under a wide variety of conditions such as pressure, heat flux, flow rate, etc., along with acoustic characteristics of the container system like resonances, transmission paths, and wall reflections, is of utmost importance [33]. An investigation by Bessho and Nishihara [34] revealed that the boiling acoustics contain more high-frequency components for larger subcooling and flow velocity conditions. In the meantime, several investigations focused on resolving the challenges in the applicability of BA in the reactors and determining the effect of various mixtures on the boiling acoustic spectra [35,36].

Furthermore, background noise elimination is tedious in the early years of research in boiling acoustics and therefore, the accuracy of the boiling prediction was questionable. Although, the complete elimination of environmental noise is impossible, there are several noise reduction tools available in signal processing. Negi et al. [37] used Audacity software tool to reduce background noise from the spectra. Ueki et al. [38] added Gaussian noise to the time series data and predicted the boiling regime with the noise added signal using a deep neural network with an accuracy of approximately 99%.

2.1. Role of power spectral density

Ponter [36] reported that, as the concentration of ethanol in the ethanol-water mixture increases, the sound intensity decreases steadily under artificial boiling conditions. Although the frequency spectra are reluctant to change with heat flux, power spectral density (PSD) shape is highly susceptible. According to Nishihara and Bessho [30], this might be due to the coalescence (as shown in Fig. 4) of the isolated bubbles in regimes of high heat flux conditions. Furthermore, the power spectral density (PSD) can also be affected by the pressure conditions of the

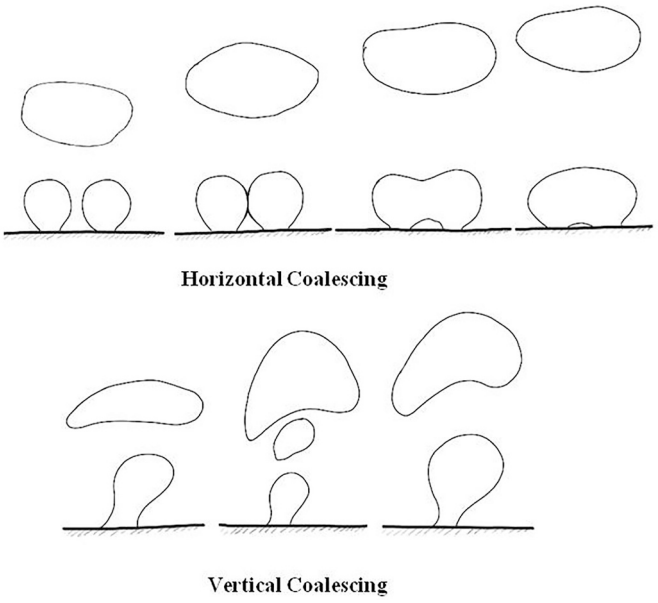


Fig. 4. Schematic diagram of coalescing phenomenon.

boiling. At higher pressures, PSD shifts towards the higher frequencies keeping the other conditions constant. This change in PSD shape concerning heat flux and system pressure may drive the research towards establishing a specific approach to detect anomalies and transitions between the boiling regimes for various applications.

Bubble diameter plays an important role in finding the dominating frequency in the boiling acoustic spectra. However, estimating the bubble diameter based on the empirical relations became quite challenging. Fig. 5 depicts the predictions of bubble diameters by various researchers. Minnaert [39] derived the natural frequency of the bubble as shown in Eq. (1).

$$f_o = \frac{1}{\pi d}\sqrt{\frac{3\gamma p_o}{\rho}} \tag{1}$$

where d is the diameter of the bubble, γ is the ratio of specific heats, p_o is the hydrodynamic pressure and ρ is the density of the liquid.

According to Minnaert [39] the bubble diameter is inversely proportional to its natural frequency. Fig. 6 shows the variation of natural frequency of the bubble with respect to the bubble diameter at various hydrostatic pressures. Likhterov [45] derived an expression given in Eq. (2) and (3) for high frequency asymptotics of acoustic spectrum taking compressibility and viscosity effects into account,

$$s(f) = \frac{\rho_\infty^{7/9} c_\infty^{17/9} R_o^{2/3}}{4\pi R^2}\left[9.72\,(p_\infty - p_\vartheta)^{20/9} f^{-10/3} + 29.5\,\mu\,(p_\infty - p_\vartheta)^{2/9} f^{-7/3}\right] \tag{2}$$

$$R_o = \frac{2\,\sigma\,T_s}{\rho_\vartheta\, l\,(T_R - T_s)} \tag{3}$$

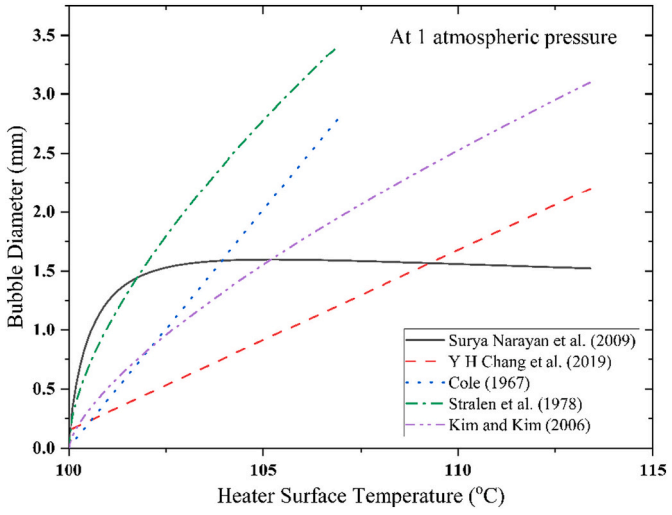


Fig. 5. Comparison of various empirical relations for bubble departure diameter [40–44].

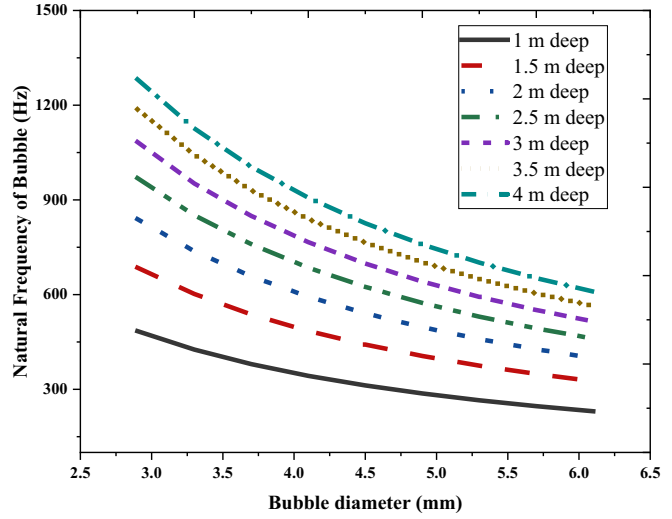


Fig. 6. Minnaert's natural frequency [39] of the vapour bubble in water at various hydrostatic pressures.

where $\rho_\infty$is liquid density unaffected by bubble, ρ_ϑ is saturated vapour density, c_∞ is sound velocity in undisturbed liquid, R_o is the equilibrium radius of bubble in liquid at uniform super heating, R is bubble radius, p_∞ is ambient liquid pressure undisturbed by the bubble, p_ϑ is the saturated vapour pressure, f is frequency, μ is dynamic viscosity coefficient, σ is surface tension, T_s is saturated temperature of vapour, T_R is instantaneous temperature at bubble boundary and l is the latent heat of vaporization.

In the investigations of Kudo et al. [46] involving the detection of boiling noise in the specific PSD region, a linear discriminant function introduced by Shimura [47] has been used and reported an excellent accuracy between the predicted and obtained results in that frequency ranges. It signifies the significant role of variation of PSD in boiling noise detection. As MacLeod [48] suggested, though the simple processing is sufficient for the detection, advanced computerized models are needed for the data processing to boost the sensitivity of the alarming systems of boiling anomaly. It leads to several experimental and analytical investigations into the various signal processing studies in boiling acoustic detection [24,45,49,50]. From the analytical analysis of acoustics in compressible fluids [49] the reported PSD is inversely proportional to the 2/3rd power of wall superheat, citing that the method involves several assumptions that may lead to considerable error.

The majority of the abovementioned research works focus on the BA for the application of detecting boiling in non-boiling reactors. Although there are wide range of applications, very few have used BA for boiling regime identification.

2.2. Regime identification

In the late 1990s, researchers focused on the fundamentals of boiling acoustics to extend its practical applicability to a wide range of applications which led to several notable works [26,51–54]. Celata et al. [26] extended the use of BA to detect specific regimes of boiling up to the critical heat flux in the fusion-type reactor cooling systems to avoid dry-out conditions. Upholding the fundamental research on BA in the early 2000s, many researchers [55–57] successfully formulated wave equations for acoustics, admitting the profound effects of capillarity, thermal, and phase change for bubbly flow in a porous medium. Yet, understanding bubble dynamics remains the most crucial step in unravelling the complex boiling phenomenon. To accomplish this, a straightforward photometric method [23] of recording the linear dimension of the bubble was in practice as it is comparably easy. Still, it is confined only to the bubbles of spherical shape. In such a technological crisis, Dorofeev and Volkova [58,59] estimated and validated the growth and collapse of vapour bubbles by assessing the volume of each bubble using the acoustic data based on the relation that gives the sound pressure generated by the gas bubble in a liquid environment as given in Eq.(4).

$$\frac{p}{\rho'} = \frac{R^2\ddot{R} + 2R\dot{R}^2}{r} - \frac{R^4\dot{R}^2}{2r^4} \tag{4}$$

where ρ' is the liquid density, R is radius of bubble and r is distance measured from bubble centre to measurement point.

After Celata et al. [26] who extended the application of boiling acoustics from boiling onset detection to the boiling regime specification, Lykov and Sinetskaya [60] carried out an outstanding experimental investigation involving the effect of boiling regimes on the boiling acoustics and its hysteresis at different boiling conditions up to the critical heat flux. This particular research work answered many questions about the credibility of acoustics in detecting the specific boiling regimes. Similarly, Okumiya et al. [61] differentiated different boiling regimes during the quenching process. They found that the characteristic frequency band is critical in determining the transition from the nucleate boiling regime to film boiling or transition boiling regime. The observed peaks in the frequency band for the nucleate boiling regime facilitated the applicability of BA in boiling regime prediction. Besides the regime detection, the mechanical properties of the metal after quenching can be directly correlated with acoustics, as demonstrated by Mojškerc et al. [62]. By the time BA was introduced, the methods for detecting boiling in the non-boiling background had been well established [63,64]. In many practical applications of boiling, the phenomenon of boiling is not visible. Hence the simple photographic analysis [23] will fail, wherein BA plays a vital role as it does not require optical visualization to detect the boiling.

The prominent investigation of Baek et al. [25,65] observed that BA at standard temperature and pressure conditions correlated well with the BA observed at very high pressure and temperature. Several investigations [25,26,60,61,65] proved the potential of BA to distinguish between different boiling regimes. Jo and Jo [66] extended the use of BA for observing triple-layer phenomenon and condensation of the vapour bubble in the subcooled boiling condition. The research work of Rapisarda et al. [67] involves the detection of boiling in a prototype beam accelerator using the acoustics detected by the hydrophones. Due to the complications involved in using the hydrophones, Arranz et al. [63] used accelerometers to correlate the results. The disadvantages associated with hydrophones are as follows: hydrophone requires penetrating

the wall to keep it inside the boiling chamber, hydrophones are sensitive to noise and position inside the boiling chamber. Details of instruments used by several authors for acoustic measurement are detailed in Table 2.

In nucleate boiling, the sound pressure level variation with the wall superheat shows N-shaped variation as shown in Fig. 7. With an increase in wall superheat, the acoustic energy shifts to lower frequency levels. In the microbubble emission boiling, the acoustic energy is steeply raised at significantly lower frequencies (<800 Hz), as observed in the investigation of Tang et al. [71].

The design and construction of the experimental setup for these investigations is simple and Fig. 8 shows simple schematic representing the essential parts. Alhashan et al. [74] also used BA to detect boiling inception in a pool boiling arrangement and to measure the liquid level throughout the experiment.

A similar investigation by Singh and Mohanty [75] estimated a 98.8% accuracy in estimating the liquid level height using acoustics compared to conventional methods. In the recent times, a few authors focused on predicting the boiling regime and developing autonomous control systems to take early emergency protocols to avoid power plant system failures.

Sinha et al. [73] reported the acoustic frequency jump from the nucleate boiling regime to the critical heat flux boiling regime and used such property to trigger the autonomous mechanism to bring back the boiling regime to nucleate boiling by appropriate measures. They also observed that the boiling noise frequency is dominant in the range of 200–250 Hz in the nucleate boiling regime, whereas, at the critical heat flux, the frequency reached about 400–500 Hz. A similar frequency shift in the boiling regimes was also confirmed by Seo and Bang [76], who noted that both low frequency and high-frequency components are present in transition boiling.

Both Sinha et al. [73] and Seo and Bang [76] have demonstrated that the BA characteristics correlate with the parameters of boiling dynamics. Moreover, simultaneous observation of BA, photographic, and thermal measurements will better explain the boiling phenomenon and emphasize the importance of acoustic data in the early detection of regime change [77]. The measurement of vibrations induced on the transfer line of cryogenic liquids or the flow boiling liquid can be used to find its flow regime and multiphase properties [78]. Involving deep learning strategies for predicting boiling heat transfer regimes could be the foreseeable future of emergency systems. The BA coupled with convolutional neural networks demonstrated >99% validity in predicting the boiling heat transfer regime [79]. Research progress in the previous years is clearly detailed in Table 3 according to the chronological order in decade intervals.

It is evident from the above discussions that the bubble characteristics are distinct among the various boiling regimes as shown in Fig. 9. In nucleate boiling, discrete bubbles will be formed on the surface as shown in Fig. 9. The bubble diameter increases with the heat flux. The bubble departure characteristics are similar within the nucleate boiling regime except the departure diameter. Near CHF, the boiling becomes rapid and bubbles begin to merge together. This vertical merging leads

Table 2
Acoustic measurement devices used by several authors.

Author	Instrumentation used	Reference
Osborne and Holland (1947)	Hydrophone	[68]
Colomb and Binford (1962)	Hydrophone	[69]
Schwartz and Siler (1965)	Hydrophone	[70]
MacLeod (1966)	Hydrophone type transducer	[35]
Nishihara and Bessho (1977)	Hydrophone	[30]
Tang et al. (2004)	Microphone	[71]
Lee and Higgs (2006)	Hydrophone	[72]
Geraldo et al. (2014)	Microphone	[64]
Sinha et al. (2019)	Hydrophone	[73]
Negi et al. (2021)	Microphone	[37]

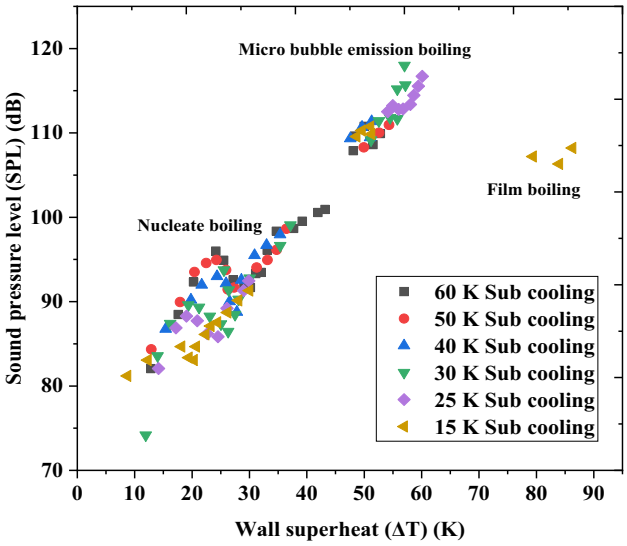


Fig. 7. N- Shaped variation in nucleate boiling regime [71].

to the mushroom shaped bubbles. A further rise in heat flux increases the coalescence of the bubbles rapidly causing vapour columns [41,88]. Moreover, bubble dynamics also varies according to the initial condition of boiling liquid, i.e., subcooled or saturated. Thus, the possible boiling regimes that need evaluation for the boiling noise characterization are 1. Subcooled and saturated boiling, 2. Nucleate boiling, 3. Critical Heat flux zone, 4. Transition and Film boiling.

2.2.1. *Subcooled and saturated boiling*

If the bulk fluid temperature is below the saturation temperature of the liquid, such condition is known to be subcooled boiling. In subcooled boiling, as the bubble rises to the free space its size reduces gradually due to the condensation of vapour on the triple line of the bubble. According to Minnaert [39] the frequency of oscillation is inversely proportional to bubble diameter. This change in frequency distribution between various subcooling levels and saturated boiling conditions is observed by Tang et al. [71]. As the bubble departs from the heater surface, encounters a bulk fluid mass of subcooled liquid. Thus, the bubble growth rate reduces drastically, and the pressure fluctuation decreases at that early stage of bubble growth. On the other hand, several investigations noted that acoustic signal amplitude in the saturated boiling condition increases with the increase in heat flux.

2.2.2. *Nucleate boiling regime*

The nucleate boiling regime starts from the formation of isolated bubbles and extends to just before the appearance of vapour columns/jets. In this regime, boiling begins with construction of tiny individual isolated bubbles at lower heat flux. As the heat flux rises, the bubble diameter at the time of departure increases. According to Sinha et al. [73], a gradual increase in the acoustic amplitude was found in the nucleate boiling regime. Several researchers, including Aoki and Welty [83], Westwater [80], Schwartz et al. [70], and Ueki et al. [38], confirmed the same for various boiling liquids. The dominating frequency in the nucleate boiling for water is 200 to 250 Hz, as quoted by Sinha et al. [73]. The dominating frequency reported in several investigations is quite different from each other. It is mainly due to the dependency of frequency distribution on the material and size of the heater [68]. Besides, it is reported by a few researchers [82,86] that coalescence in bubbles have a higher amplitude compared to others. Fig. 10 shows the comparison of frequency spectra observed by Sinha et al. [79] and Negi et al. [37].

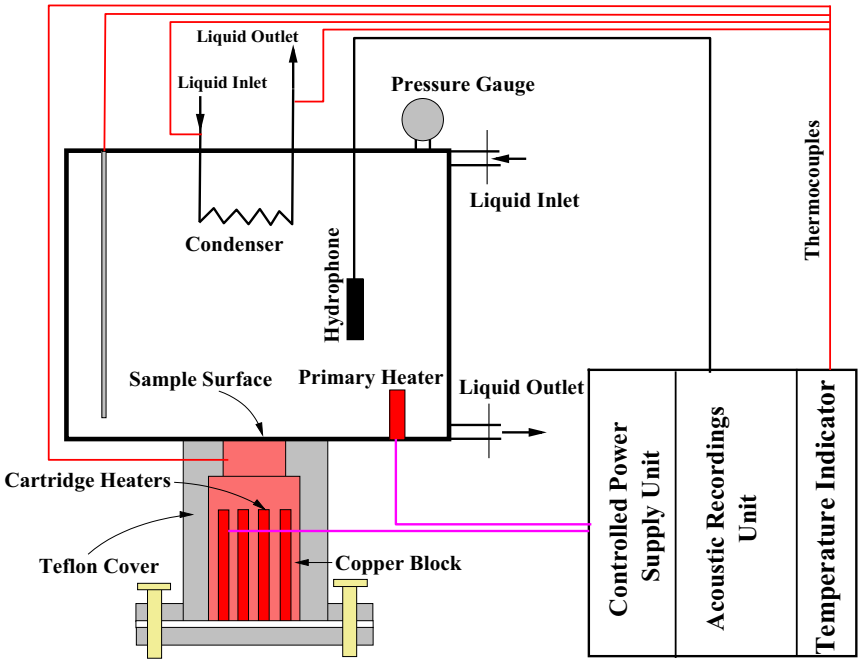


Fig. 8. Simple schematic of experimental setup.

Table 3
Research progress in boiling acoustics through the decades.

Year	Key take away	References
Till 1970	• Recommendations for the advancements for the safety systems in the power plants. • Acoustic methods are suggested for detection of boiling incipience in water cooled reactors. • Bubble volume resonances are theoretically determined. • The acoustic spectrum of boiling is reported to be random noise appearance. • Suggested quantitative studies in boiling detection.	[39,47,68–70,80–83]
1970–1980	• More studies using hydrophones have been investigated to understand the boiling acoustics including the triangularization studies to locate the anomalous regions. • Resonance peaks observed to be the obstacles for the proper triangularization of boiling acoustic signals. • Suggested for more studies involving effects of sub-cooling, operating pressure, flow rate and heat flux. • Photographic study of bubble dynamics and acoustic analysis have been studied simultaneously to found the prominent effect of bubble diameter on acoustic spectrum. • Linear decrement functions and probability density functions were established for the detection of boiling incipience.	[28–36]
1980–1990	• Dominant frequency range was found to 200–500 Hz. • Several signal processing techniques were established for boiling acoustic study. • Boiling acoustics for mixtures of fluid were investigated but no solid conclusions have been brought out. • Strengthening the basic principles research has been dominant in investigating and understanding the boiling spectra which brought out several notable conclusions which makes the first ever step to unravel the potentiality of boiling acoustics on differentiating the boiling regimes. • Research on the detection of boiling incipience has been still continued for the its applicability in non – boiling reactors	[24,45,46,48,50]
1990–2000	• Many investigations related to the elimination of background noise and electrical interference that are hampering the analysis on acoustic spectra have been carried out. • Boiling noise is highly dependent on the compressibility and viscosity of the fluids. • Very few investigations confirmed the potentiality of the boiling acoustics for its ability to differentiate between the different regimes of boiling.	[23,26,49,51–53]
2000–2010	• Investigations on the effect of bubble parameters including bubble coalescence and collapse on boiling acoustics have been reported. • Several investigations to estimate the dynamics of vapour bubble volume variation has been reported. • Acoustic techniques have been developed to discriminate the boiling regime in a quenching process.	[55,58,59,61,72,84–86]
2010 - Present	• Detailed research has been carried out on boiling detection and artificial neural networks have been developed to detect the boiling incipience leading a way to autonomous cooling systems. • Artificial neural networks and convolutional neural networks has been dominant to predict the boiling regime through boiling acoustics. • Several methods have been proposed to detect CHF using boiling acoustics.	[25,37,38,56,57,62–64,66,67,71,73,74,77–79,87]

2.2.3. Critical heat flux

In power plants and cooling infrastructure systems, detecting critical heat flux point plays a vital role in assessing the plant safety operation. It is due to film boiling, which happens just after the CHF point. Detection of CHF using the heat transfer parameters is a theoretical method with several disadvantages. Of late, acoustic methods are getting popular due

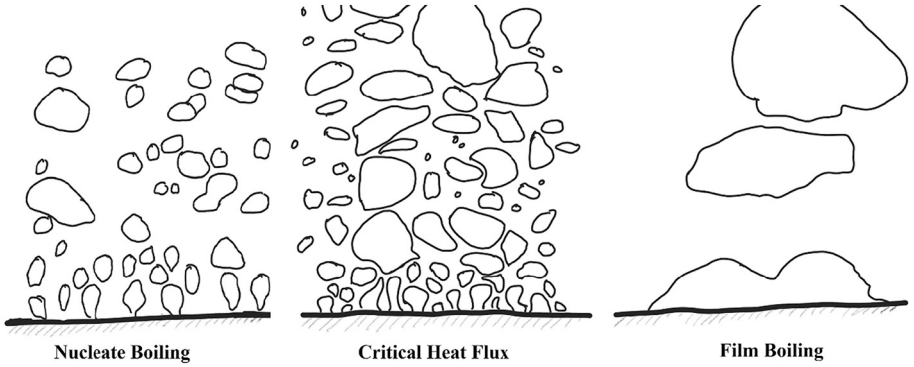


Fig. 9. Bubble dynamics at various boiling regimes.

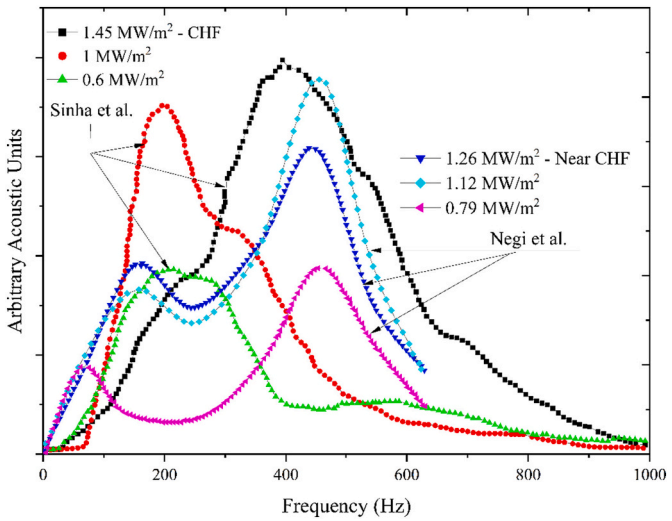


Fig. 10. Comparison of frequency spectra observed by Sinha et al. [79] and Negi et al. [37].

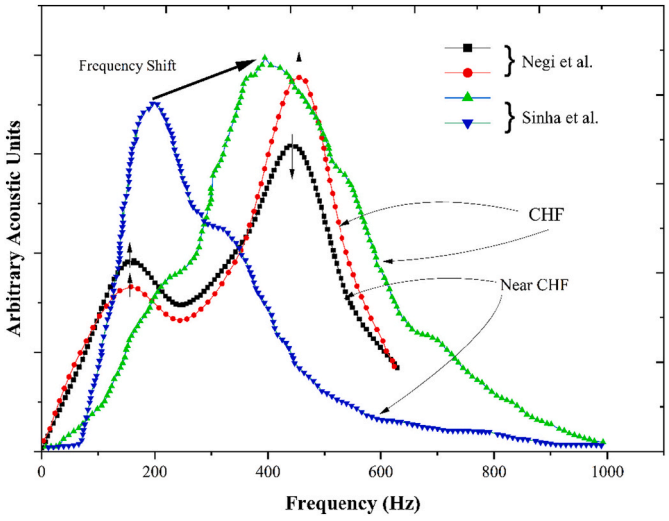


Fig. 12. CHF identification observed by Negi et al. [37] compared to Sinha et al. [74].

to their accuracy and real-time analysis. Sinha et al. [73] found a sudden frequency shift from 200 to 250 Hz to 400–500 Hz at the CHF. It is due to the change of bubble shape from mushroom-shaped to smaller and individual bubbles as shown in Fig. 11. Aoki and Welty [83] found a sudden sound attenuation just after CHF, which might be due to the vapour layer formation. However, Westwater [80] did not observe any drop in SPL. Alternatively, the SPL exhibits an increasing trend with heat flux in both the nucleate boiling region and transition boiling region.

Negi et al. [37] found a drop in SPL near CHF and suggested it could be the potential trigger to identify the CHF. Moreover, Negi et al. [37] found an increased amplitude in the lower frequency peaks and a decreasing trend in the dominating frequency trend as shown in Fig. 12.

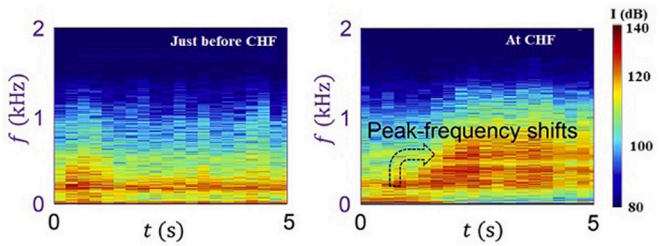


Fig. 11. CHF identification observed by Sinha et al. [79].

2.2.4. Transition and film boiling

If the heat flux is raised even after the CHF, then bubbles will merge rapidly into each other and form vapour columns; this state of boiling is commonly identified as transition boiling. A further increase in the heat flux creates a vapour layer on the heater surface, forming a vapour blanket on top of the heater surface. It prevents the heat transfer from the surface of the heater resulting in lowest heat transfer. The regime of vapour blanket is called film boiling. Many studies focused on the CHF, and yet very few studies have reported the acoustic characterization in the film boiling regime. According to Tang et al. [71] there is a steep rise in acoustic pressure in the transition boiling regime due to the oscillation of vapour film. The bulk volumetric fluctuation of fluid caused a steep rise in acoustic pressure in the transition and film boiling region; however, no microbubbles are present in the film boiling regime. The dominating frequency in these regimes is reported as 200–800 Hz. In addition, the low-frequency range's energy is more significant than nucleate boiling [61]. Aoki and Welty [83], Grum et al. [62], and Westwater [80] obtained similar results of rise in acoustic pressure.

2.3. Significance of ANN in BA

Deriving a direct correlation among all the researchers' data and regimenting the terms is beyond human possibility due to the error-bars specified in reported data and intricacy in merging diverse conditions. Hence, in recent times many of the researchers are moving towards Artificial Neural Networks (ANN). Researchers utilized the ANN tool to predict the boiling heat transfer by giving voltage and thermo-physical

conditions of the heater as well as boiling liquid as inputs and heat transfer rate, heat transfer coefficient, and wall superheat as output [89–92]. This method of employing worked remarkably fine until its real-time usage. The heat transfer parameters such as temperature rise rapidly during emergency conditions, providing no room for control measures. Such conditions could potentially make the safety systems questionable. Though the systems with stringent rules and regulations are the safest, human error makes a big difference. In the Chernobyl disaster, shutting down of the cooling system led to the most devastating incident causing a massive mortality which involved human error partly [93]. The most efficient way of predicting the unwanted heat transfer reduction in such cooling systems is to intuitively understand the boiling curve and its regimes.

3. Conclusion

This paper reviews the most classical works of literature on boiling acoustics. Most important observations are summarized as follows. BA has high-frequency components for larger sub-cooling and in the flow velocity conditions. The oscillating frequency of the bubble is inversely proportional to the bubble diameter. Volume oscillation while coalescence initiates a pressure-equalizing mechanism, which is the leading cause of sound. When bubble generation is rapid, bubble coalescence becomes dominant in sound. The sound of boiling mainly depends on bubble coalescence, growth, oscillations during departure, and bubble collapse. In contrast, the frequency distribution primarily depends on the material and size of the boiling surface. In liquid mixtures, the sound intensity of the BA is highly dependent on the concentration of mixtures. The frequency spectra are reluctant to change within a boiling regime, whereas, the Power Spectral Density is sensitive to the heat flux change. It has been reported in several investigations that BA is not only credible to detect the boiling inception but also has the potential to detect different regimes of the boiling. Though the hydrophones and accelerometers serve a similar purpose in BA, accelerometers are frequently used due to their easy installation.

Few researchers' reported “N” shaped variation in a plot of sound pressure level and wall superheat in sub-cooling conditions. Moreover, when the wall superheat increases, the acoustic energy shifts towards lower frequency levels. BA is predominant between 200 and 250 Hz in the nucleate boiling regime and 400–500 Hz in the critical frequency regime. A sudden drop in acoustic pressure just before CHF is mainly attributed to the bubble shape change from mushroom-shaped to tiny and individual bubbles.

It is evident from the reported literature that the BA can be influenced by thermo-physical properties of the boiling medium, heater surface and other enhancement strategies like surface modifications that results in change in bubble dynamics. Till date the available data to completely understand the BA pertaining to practical enhancement strategies is very limited. Hence, the research towards such investigations should be encouraged to implement the BA based autonomous cooling systems.

Credit author statement

Sreeram Barathula: Literature Survey, Writing – original draft, Writing – review & editing.

K. Srinivasan: Conceptualization, Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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